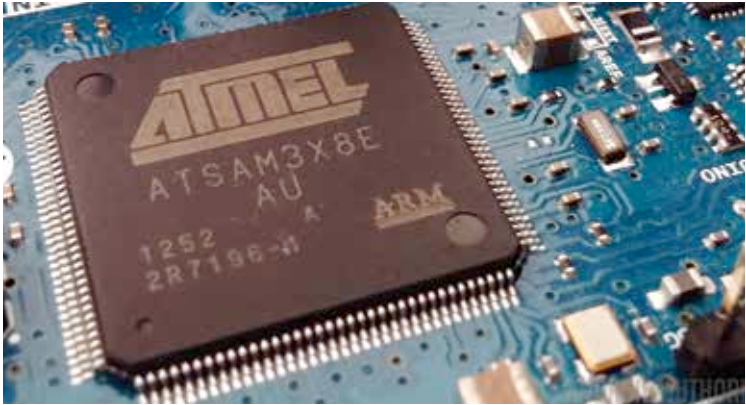


several other peripheral circuits and memory. The tendency of the tech industry to make gadgets more compact has not been different in the case of computers. Since the time of Charles Babbage, where a ‘computer’ occupied the whole size of a room, we’ve come to a time of nanobot computers that are microscopic. We shall look at how the first system on a chip came about to be and its evolution.



A widely used ARM microcontroller

A Watch Computer

SoC is a pretty wide term as it is the name given to any circuit which integrates electronic components. However, the first major breakthrough in this field was seen in the 1970s when Electro-Data Inc. released the ‘Wrist computer’ (which was basically a digital watch) designed by George Thies and Willy Crabtree. What the designers did was reduce the 44 chip and 4,000 bonding requirement for creating a timekeeping device to a single chip which integrated an LCD display with timekeeping circuitry. This was a pretty bizarre concept for the time and the watch was bizarrely priced at \$2,000 which was a hole in your pocket for the time.

The first true SoC, however, appeared in a Microma watch in 1974. The watch was designed in such a way that it integrated LCD driver transistors as well as the timing functions to a single INTEL 5810 CMOS chip. TI released a single chip LCD watch in 1976 priced just at 20\$. This was a big move and a huge game changer for the timekeeping industry. It was quickly realized that the commercial challenge with the use of SoCs lies in making the tech as affordable as possible. Timex and a dozen other companies entered the